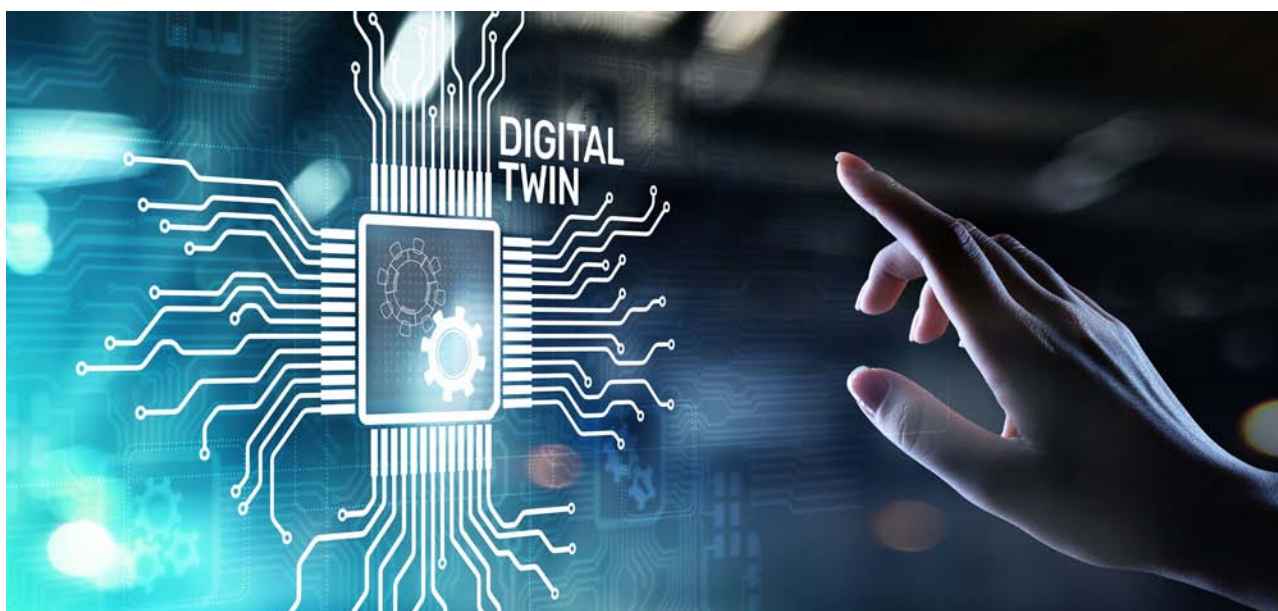


What Digital Twins Bring to the Metaverse

The metaverse, where real life converges with the digital world, has given birth to many new age business models and use cases. Digital twins within the metaverse offer new ways of interacting with the real world. Let's see how...



One of the key technologies driving the evolution of the metaverse is that of digital twins. Most popular in the retail industry, a digital twin is a virtual replica of a physical object or system, offering a comprehensive digital representation that can be used for various purposes. When combined with the metaverse, digital twins can enable new ways of experiencing and interacting with the real world.

Architecture of metaverse-based digital twins

The architecture of a metaverse-based digital twin consists of several layers. At the bottom layer is the physical asset or system that the digital twin is replicating. This could be anything from a building, to a manufacturing plant, to

a vehicle. The physical asset is equipped with sensors and other IoT devices that collect data on its operations, condition, and performance. This data is then transmitted to the digital twin, which is hosted in the cloud or on-premises.

The next layer is the data processing layer, which consists of several components that work together to process the data collected from the physical asset. This layer typically includes data ingestion and storage, data processing and analysis, and data visualisation and reporting. Data ingestion and storage involves collecting data from various sources, and storing it in a scalable and secure data store. Data processing and analysis involves applying advanced analytics techniques, such as machine learning and artificial intelligence, to the collected data to

extract insights and identify patterns. Data visualisation and reporting involves presenting the results of the analysis in an easy-to-understand format.

The third layer is the digital twin layer, which is responsible for creating a virtual representation of the physical asset or system. The digital twin layer consists of several components, including the virtualization engine, the digital twin model, and the digital twin APIs. The virtualization engine is responsible for creating and managing the digital twin, while the digital twin model defines the relationships between the physical asset and its virtual counterpart. The digital twin APIs provide an interface for interacting with the digital twin, enabling users to view and manipulate the virtual representation of the physical asset.